

MLPerf Storage v3.0 — System Description

potato1-qlc-closed-tcm · CLOSED division · submitter: farmgpu

System name	potato1-qlc-closed-tcm
Submitter	FarmGPU (farmgpu.com)
Division	CLOSED
Benchmark	KVCache — MLPerf Storage v3.0 (mlpstorage 3.0.37)
Solution type	local-storage (direct-attached / integrated client storage)
Availability	available
Usable capacity	2682.7248 TiB
Deployment	on-premises, bare metal
Rack units	0 U storage-under-test (integrated in client nodes; clients excluded per the MLPerf RU definition). Physical deployment footprint: 1 x 2U = 2 U of client nodes.
Client topology	1 node(s) x 1 MPI rank(s)/node = 1 client rank(s)
Model(s) exercised	llama3.1-8b (Opt 1, 2); llama3.1-70b-instruct (Opt 3)
Run date(s)	2026-07-08 / 2026-07-09
Document prepared	2026-08-03

1. System overview

The storage under test is client-local NVMe: 1 identical bare-metal node(s), each benchmarking its own independent 24-drive Solidigm™ D5-P5336 QLC array (md RAID-0, 512 KiB chunk, XFS at /mnt/qlc). There is no external or shared storage system and no data sharing between nodes; the 400GbE fabric carries only MPI coordination and result-file aggregation, not benchmark I/O. Aggregate results scale with client count (Rules §6.3.3). Measured usable capacity is 2682.7248 TiB across 1 node(s), taken from the mounted filesystem rather than computed from drive counts.

2. Hardware

Chassis	Supermicro SYS-212H-TN (2U)
CPU	1x Intel® Xeon® 6781P (80 cores / 160 threads)
DRAM	512 GiB installed — 8 x 64 GB DDR5-5600 (Micron MTC36F2046S1PC56BG1); 503 GiB available to the OS
Storage under test	24x Solidigm™ D5-P5336 122.88 TB QLC PCIe Gen4 NVMe (P/N SBFPF2BV0P12); single md RAID-0 (512 KiB chunk) + XFS, mounted at /mnt/qlc
Data NIC	2 × NVIDIA® ConnectX®-7 400GbE (RoCE-capable)
Management NIC	1 × 1 GbE
Power supplies	2x 1600 W Titanium (208 V); min PSUs active = 1 (1 redundant)

3. Software

Operating system	Ubuntu 24.04
Benchmark	MLCommons mlpstorage 3.0.37, kv_cache workload
Storage stack	Linux md RAID-0 (512 KiB chunk) + XFS
Launcher	Open MPI (mpirun)
Memory allocator	tcmalloc (libtcmalloc_minimal.so.4)

4. Settings and non-default configuration

- tcmalloc (libtcmalloc_minimal.so.4) as the client-side memory allocator, injected via LD_PRELOAD through mpirun (--mpi-params). This is the only client tuning applied.
- Single-host run: all harness pre-run probes were active; --skip-validation was NOT used.

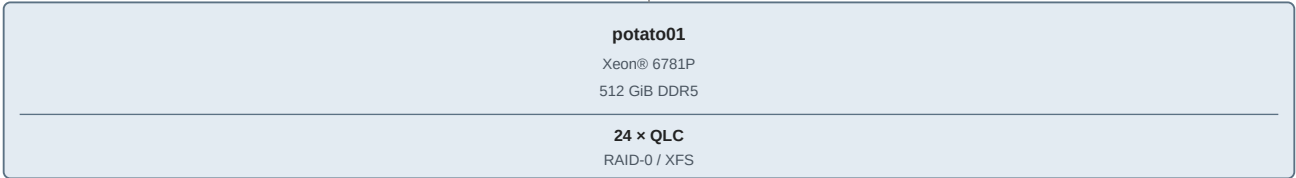
5. Power requirements

System component	Power supply unit	Nameplate rated power	Design power
Client/storage node x1	PSU 1	1600 W	1600 W
	PSU 2 (redundant)	1600 W	—
Totals		1 x 3200 W	1 x 1600 W

Design power is stated at one active PSU per node (redundant configuration, min PSUs active = 1); nameplate totals count both PSUs. Client-network switching is excluded per §11.4.2. Measured wall power was not captured; values are nameplate-derived and represent provisioned rather than consumed power.

6. System topology

400 GbE — MPI rank coordination and results aggregation only · NO benchmark I/O on the wire



STORAGE UNDER TEST — shaded region

24 × Solidigm™ D5-P5336 122.88 TB QLC NVMe, one independent md RAID-0 + XFS array per node at /mnt/qlc · 1 × 1 = 1 MPI rank(s)

Each node is self-contained; no data is shared between nodes and no storage traffic crosses the fabric.

Each of the 1 node(s) is self-contained: CPU + 512 GiB installed DDR5 + a local 24x QLC md-RAID0/XFS array (the storage under test). No storage traffic crosses the fabric; the 400GbE links carry MPI coordination and NFS result aggregation only.

7. Rack units

Per the MLPerf Storage RU definition, total_rack_units counts the storage-under-test plus any required backend gear and excludes client nodes. This solution integrates its storage into the client nodes with no dedicated storage chassis or backend switching, so total_rack_units = 0 (the systems YAML records 0, which the submission validator confirms). For physical context, the deployment occupies 1 x 2U = 2 U of client-node rack space.